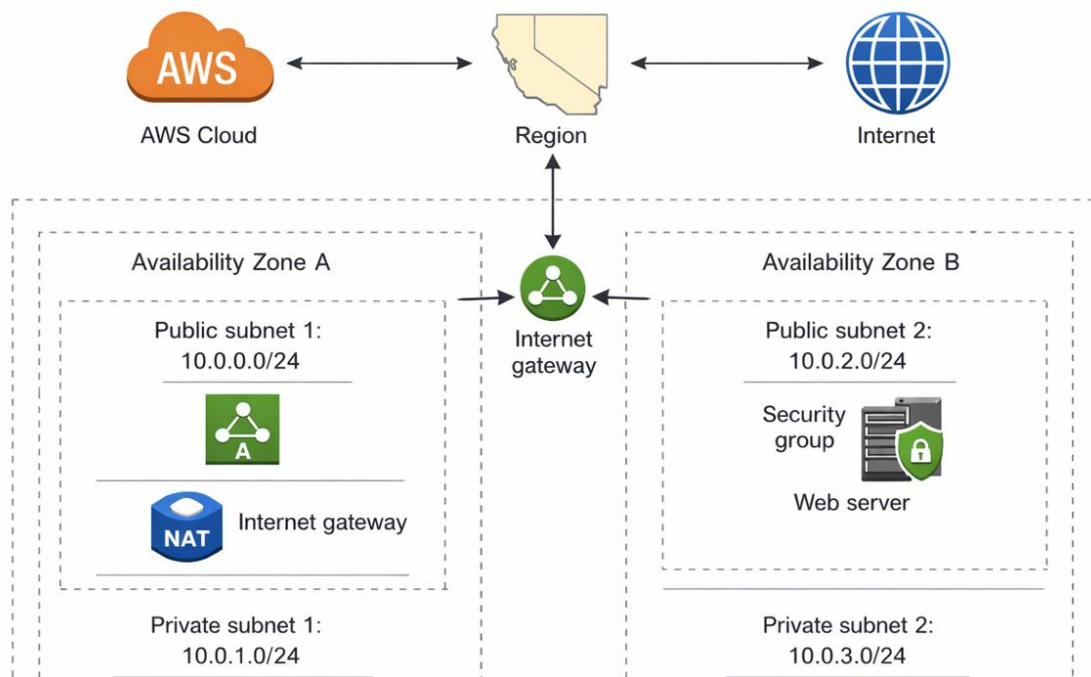
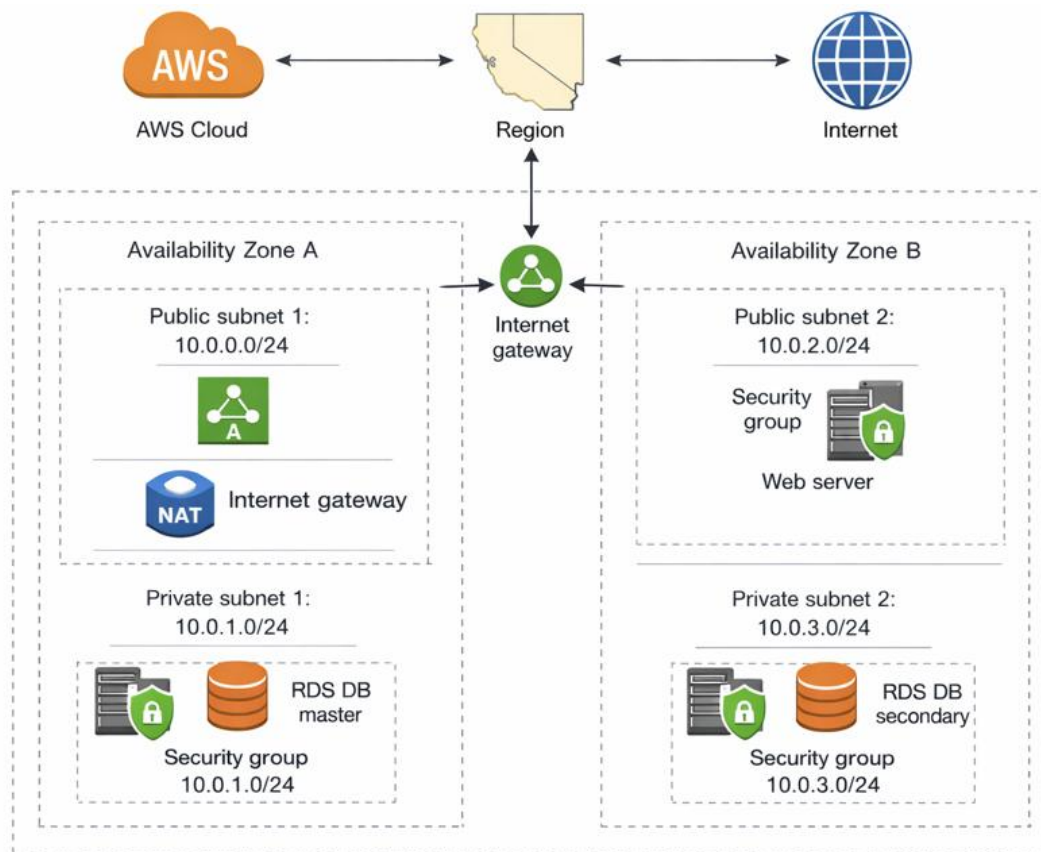


(Module 8 Lab - Build a Database Server)

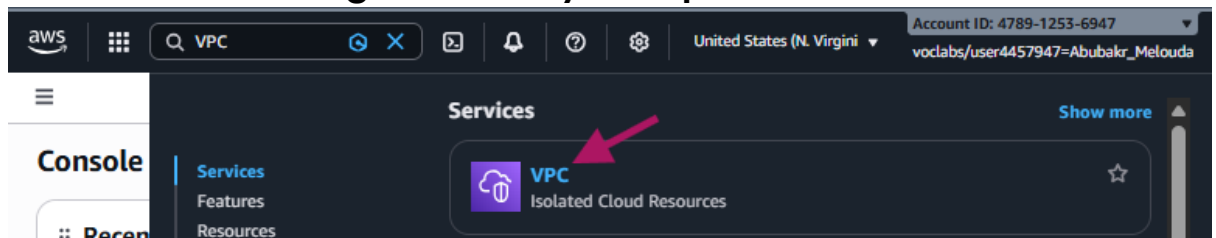
Current Lab Infrastructure diagram



Proposed Infrastructure with database server



- **Screenshot creating DB Security Group in Lab VPC**



The first screenshot shows the AWS Management Console 'Security Groups' page. The left sidebar has a red arrow pointing to 'Security groups' under the 'Security' section. The main content area shows a list of 5 security groups. A red arrow points to the 'Create security group' button in the top right.

Name	Security group ID	Security group
-	sg-0be3941f8a6b14a15	default
-	sg-0f4db1334a37d55d1	default
-	sg-0efa4df6106551c42	default
-	sg-09e5babb638921076	WorkEc2Se
Web Security Group	sg-08afa324221d390e0	Web Securi

The second screenshot shows the 'Create security group' form. The fields are highlighted with red boxes:

- Security group name:** DB Security Group
- Description:** Permit access from Web Security Group
- VPC:** vpc-0e7d1e7b6548ca833 (Lab VPC)

- Screenshot configuring inbound rule allowing MySQL from Web Security Group

aws | United States (N. Virginia) | Account ID: 4789-1253-6947 | voclabs/user4457947=Abubakr_Melouda

VPC > Security Groups > Create security group

Outbound rule 1 Delete

Type: All traffic | Protocol: All | Port range: All

Destination type: Custom | Destination: 0.0.0.0/0

Outbound rule 2 Delete

Type: MySQL/Aurora | Protocol: TCP | Port range: 3306

Destination type: Custom | Destination: sg-08afa324221d390e0

Destination dropdown:

- CIDR blocks
- Security Groups
 - Web Security Group | **sg-08afa324221d390e0**
- Prefix lists

Warning: Rules with destination of 0.0.0.0/0 or ::: recommend setting security group rules to specific known IP addresses.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

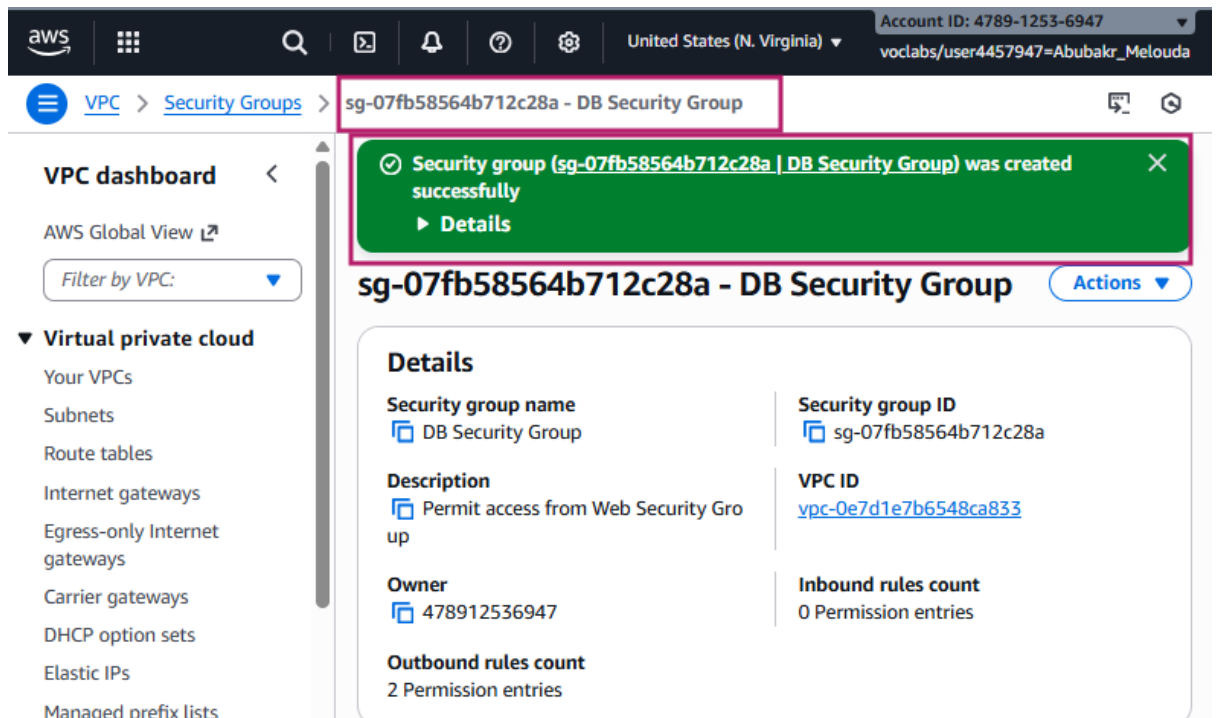
No tags associated with the resource.

Add new tag

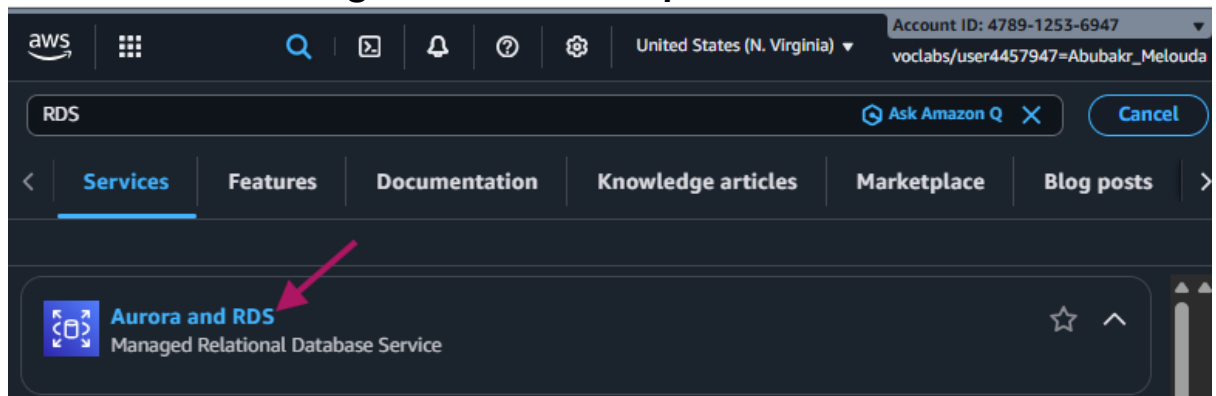
You can add up to 50 more tags

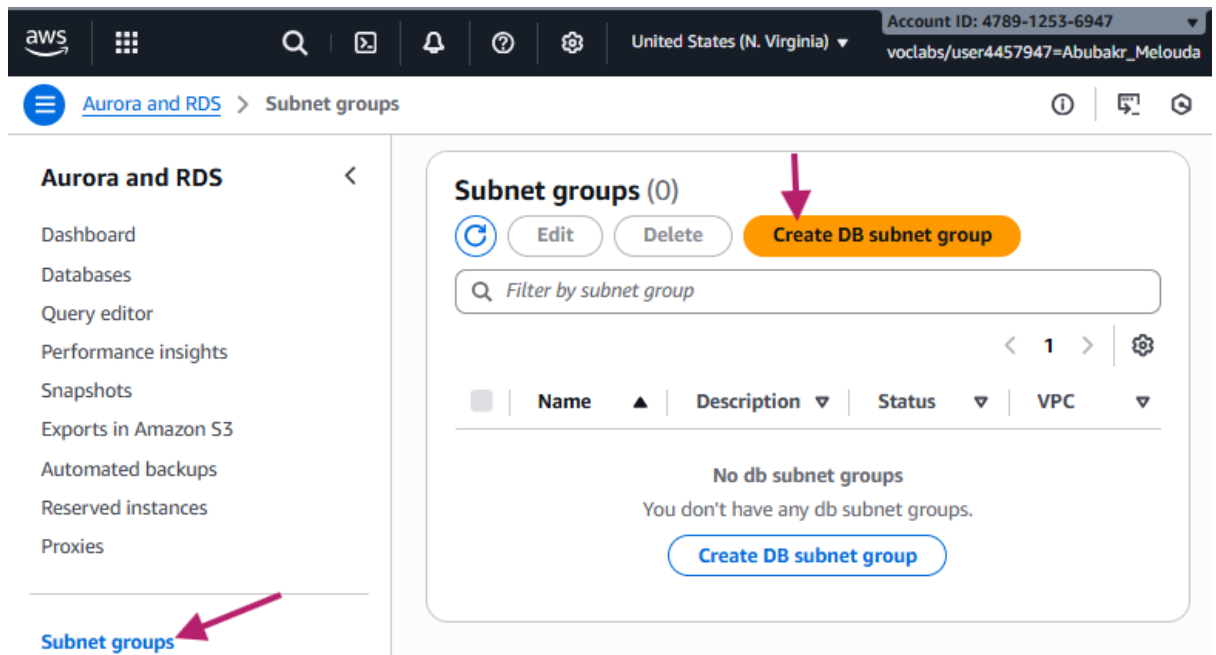
Cancel Create security group

- Screenshot DB Security Group successfully created



- Screenshot creating DB Subnet Group for RDS





- Screenshot selecting subnets in two Availability Zones for high availability

Create DB subnet group

To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.

Subnet group details

Name
You won't be able to modify the name after your subnet group has been created.
DB-Subnet-Group
Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

Description
DB Subnet Group

VPC
Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.
Lab VPC (vpc-0e7d1e7b6548ca833)
4 Subnets, 2 Availability Zones

Add subnets

Availability Zones
Choose the Availability Zones that include the subnets you want to add.
Choose an availability zone
us-east-1a X us-east-1b X

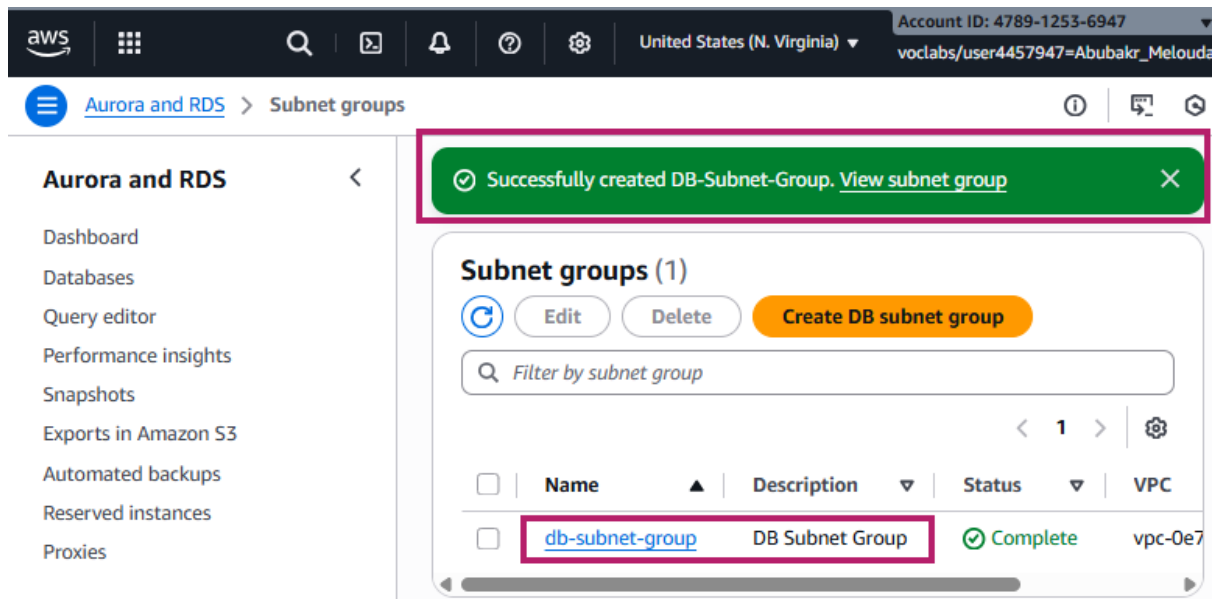
Subnets
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.
Select subnets
Private Subnet 1
Subnet ID: subnet-0d4ae81d265bfa73d CIDR: 10.0.1.0/24 X
Private Subnet 2
Subnet ID: subnet-078042ee5086a6c56 CIDR: 10.0.3.0/24 X

Subnets selected (2)

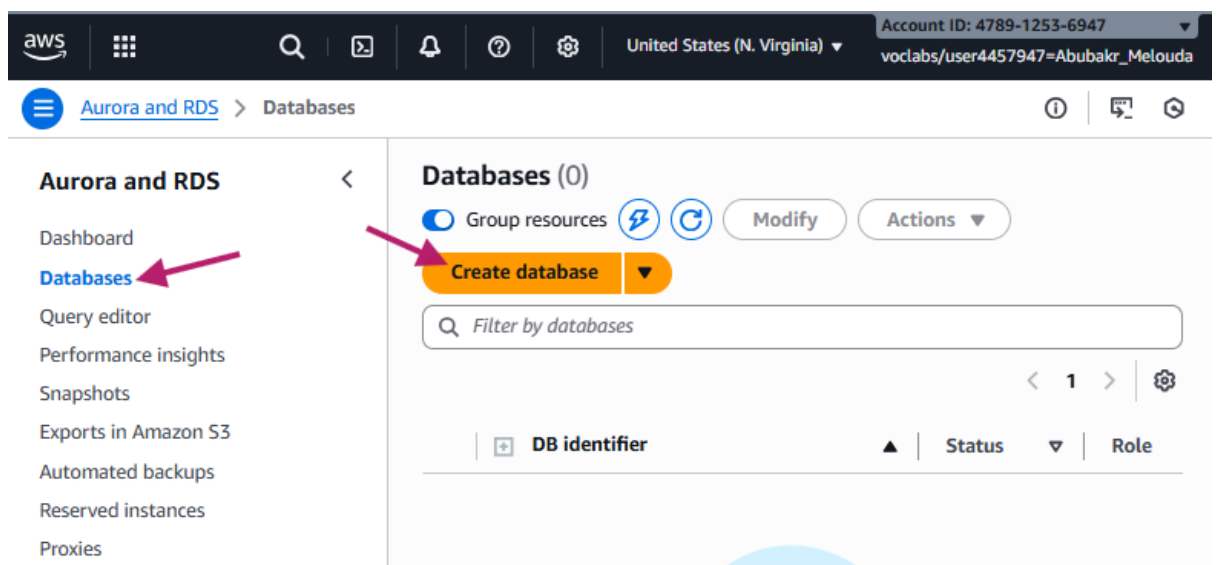
Availability zone	Subnet name	Subnet ID	CIDR block
us-east-1a	Private Subnet 1	subnet-0d4ae81d265bfa73d	10.0.1.0/24
us-east-1b	Private Subnet 2	subnet-078042ee5086a6c56	10.0.3.0/24

Cancel Create

- Screenshot DB Subnet Group successfully created



- Screenshot selecting MySQL database engine with Dev/Test template



United States (N. Virginia) ▼Account ID: 4789-1253-6947
voclabs/user4457947=Abubakr_Melouda

[Aurora and RDS](#) > [Databases](#) > Create database

Create database [Info](#)

Choose a database creation method

☒ **Full configuration**
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ **Easy create**
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible)


☐ Aurora (PostgreSQL Compatible)


☒ **MySQL**


☐ PostgreSQL


☐ MariaDB


☐ Oracle


☐ Microsoft SQL Server


☐ IBM Db2


Edition

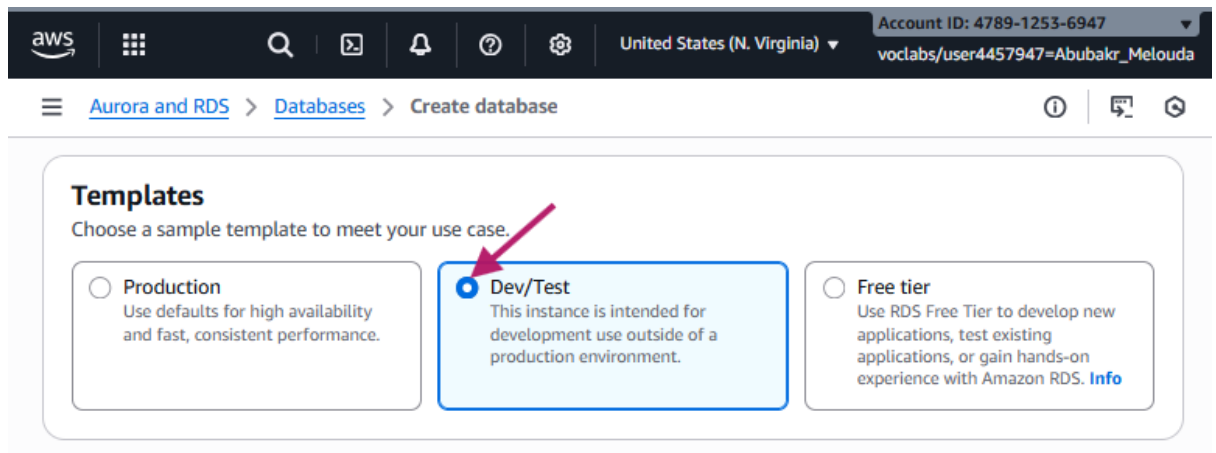
☒ **MySQL Community**

Engine version [Info](#)
View the engine versions that support the following database features.

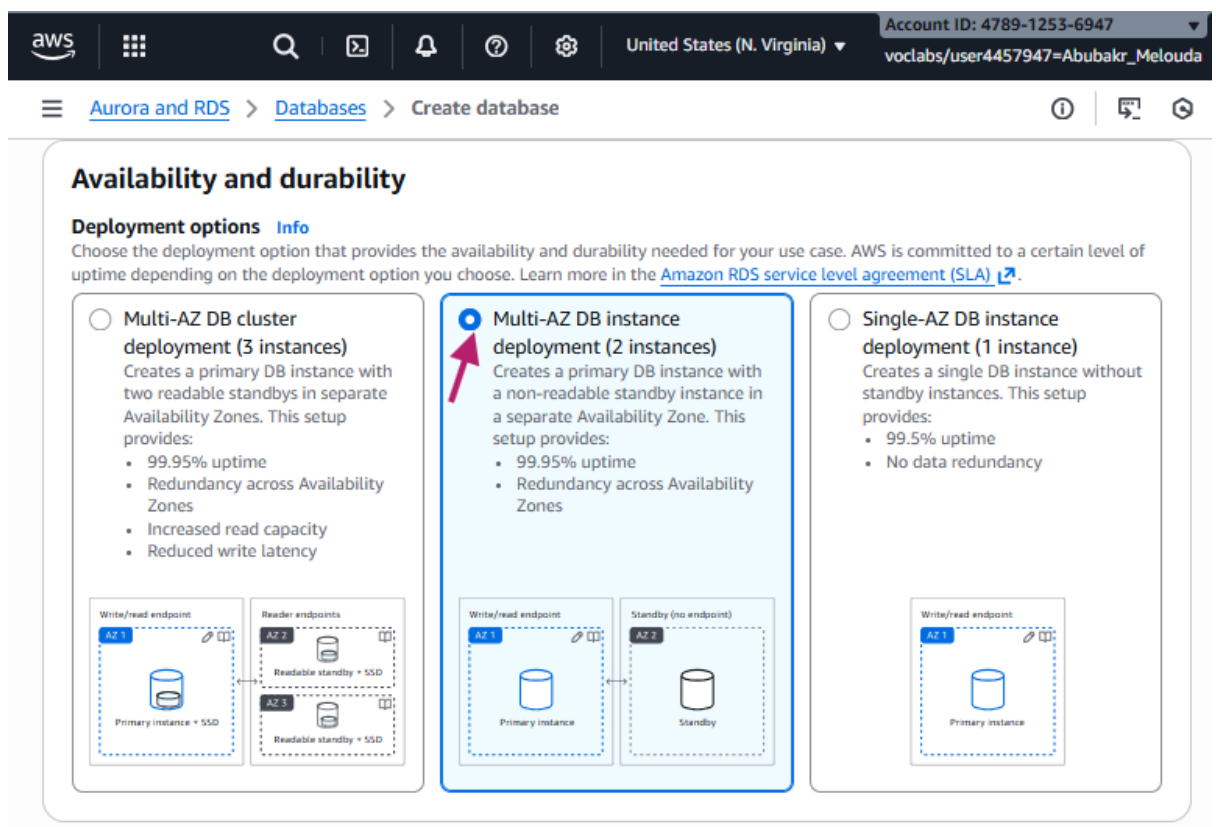
▼ **Hide filters**

☐ Show only versions that support the Multi-AZ DB cluster [Info](#)
Create a Multi-AZ DB cluster with one primary DB instance and two readable standby DB instances. Multi-AZ DB clusters provide up to 2x faster transaction commit latency and automatic failover in typically under 35 seconds.

☐ Show only versions that support the Amazon RDS Optimized Writes [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.



- Screenshot configuring Multi-AZ deployment for high availability



- Screenshot database settings and t3.micro instance class configuration

aws

United States (N. Virginia)

Account ID: 4789-1253-6947
voclabs/user4457947=Abubakr_Melouda

Aurora and RDS

Databases

Create database

Settings

DB instance identifier [Info](#)

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

lab-db

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

▼ Credentials Settings

Master username [Info](#)

Type a login ID for the master user of your DB instance.

main

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

☐ **Managed in AWS Secrets Manager - *most secure***
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ **Self managed**
Create your own password or have RDS create a password that you manage.

☐ **Auto generate password**
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

.....

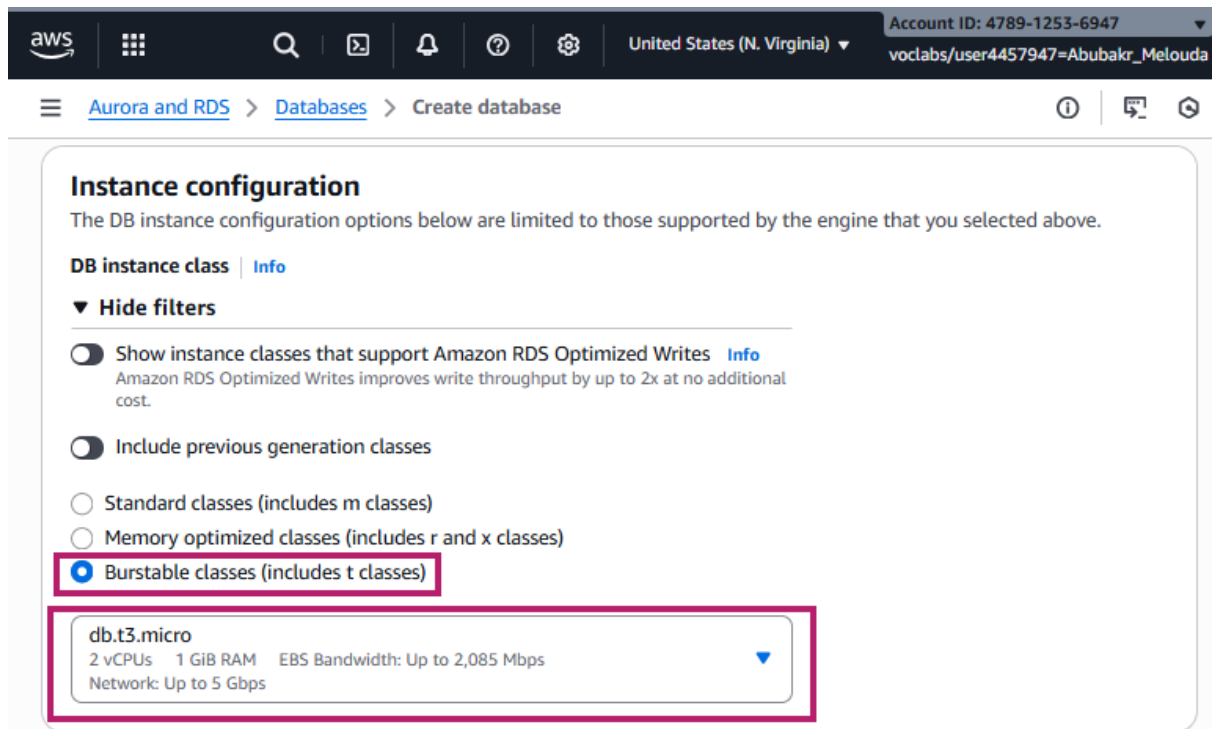
Password strength

Average

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' " @

Confirm master password [Info](#)

.....



- Screenshot storage configuration and network connectivity settings

aws

United States (N. Virginia)

Account ID: 4789-1253-6947

voclabs/user4457947=Abubakr_Melouda

Aurora and RDS

Databases

Create database

Storage

Storage type [Info](#)
Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp3)
Performance scales independently from storage

Allocated storage [Info](#)
20 GiB
Minimum: 20 GiB. Maximum: 6,144 GiB

Provisioned IOPS [Info](#)
3000 IOPS
Baseline IOPS of 3,000 IOPS is included for allocated storage less than 400 GiB.

Storage throughput [Info](#)
125 MiBps
Baseline storage throughput of 125 MiBps is included for allocated storage less than 400 GiB.

To provision additional IOPS and throughput, increase the allocated storage to 400 GiB or greater.

► Additional storage configuration

Connectivity [Info](#)

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ **Don't connect to an EC2 compute resource**
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ **Connect to an EC2 compute resource**
Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) [Info](#)
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Lab VPC (vpc-0e7d1e7b6548ca833)
4 Subnets, 2 Availability Zones

Only VPCs with a corresponding DB subnet group are listed.

DB subnet group [Info](#)
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

db-subnet-group
2 Subnets, 2 Availability Zones

Public access [Info](#)

☐ **Yes**
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☒ **No**
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ **Choose existing**
Choose existing VPC security groups

☐ **Create new**
Create new VPC security group

Existing VPC security groups

Choose one or more options

DB Security Group X

- Screenshot additional configuration enabling encryption, enhanced monitoring and automated backup

aws

United States (N. Virginia)

Account ID: 4789-1253-6947
voclabs/user4457947=Abubakr_Melouda

Aurora and RDS

Databases

Create database

▼ Additional configuration

Database options, encryption turned on, backup turned on, backtrack turned off, maintenance, CloudWatch Logs, delete protection turned off.

Database options

Initial database name [Info](#)

lab

If you do not specify a database name, Amazon RDS does not create a database.

DB parameter group [Info](#)

default.mysql8.0

Option group [Info](#)

default:mysql-8-0

Backup

☒ Enable automated backup
Creates a point-in-time snapshot of your database

⚠ Please note that automated backups are currently supported for InnoDB storage engine only. If you are using MyISAM, refer to details [here](#).

Backup retention period [Info](#)

The number of days (1-35) for which automatic backups are kept.

7 days

Backup window [Info](#)

The daily time range (in UTC) during which RDS takes automated backups.

☐ Choose a window

☒ No preference

Backup tags [Info](#)

Copies tags from the source database to the automated backup and snapshots respectively.

☐ Copy tags to automated backup
This is a one-time setting. Future tag modifications need manual updates.

☒ Copy tags to snapshots

This screenshot shows the 'Encryption' section of the AWS 'Create database' page. The 'Enable encryption' checkbox is checked. Below it, the 'AWS KMS key' dropdown menu is open, showing 'Enter a key ARN'. The 'Amazon Resource Name (ARN)' field contains the value 'arn:aws:ec2:us-east-1:478912536947:instance/i-0cc5cd136d3648de5'. An example ARN is provided: 'arn:aws:kms:<region>:<accountID>:key/<key-id>'. The account ID '478912536947' is displayed at the bottom.

Account ID: 4789-1253-6947
voclabs/user4457947=Abubakr_Melouda

Enable encryption
Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console. [Info](#)

AWS KMS key [Info](#)
Enter a key ARN

Amazon Resource Name (ARN)
arn:aws:ec2:us-east-1:478912536947:instance/i-0cc5cd136d3648de5
Example: arn:aws:kms:<region>:<accountID>:key/<key-id>

Account
478912536947

This screenshot shows the 'Monitoring' section of the AWS 'Create database' page. The 'Database Insights - Standard' option is selected. Below it, the 'Additional monitoring settings' section is expanded, showing 'Enhanced Monitoring' with the 'Enable Enhanced monitoring' checkbox checked. A red arrow points to this checkbox.

Account ID: 4789-1253-6947
voclabs/user4457947=Abubakr_Melouda

Monitoring [Info](#)
Choose monitoring tools for this database. Database Insights provides a combined view of Performance Insights and Enhanced Monitoring for your fleet of databases. **Database Insights** pricing is separate from RDS monthly estimates. See [Amazon CloudWatch pricing](#).

☐ Database Insights - Advanced

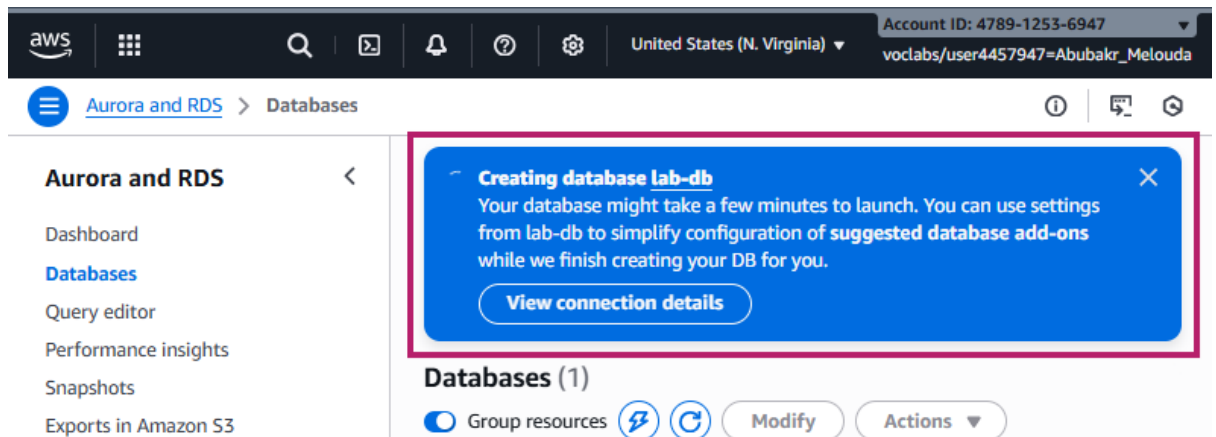
- Retains 15 months of performance history
- Fleet-level monitoring
- Integration with CloudWatch Application Signals

☒ Database Insights - Standard

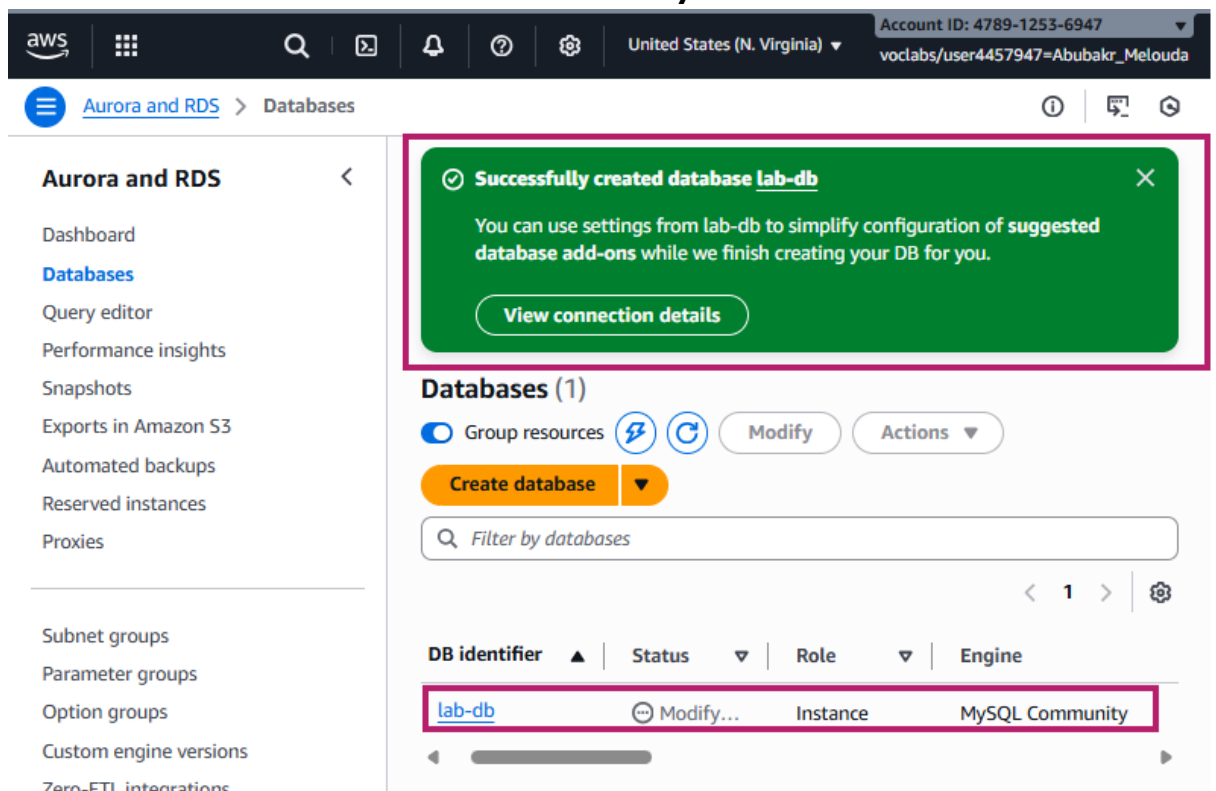
Additional monitoring settings
Enhanced Monitoring, CloudWatch Logs and DevOps Guru

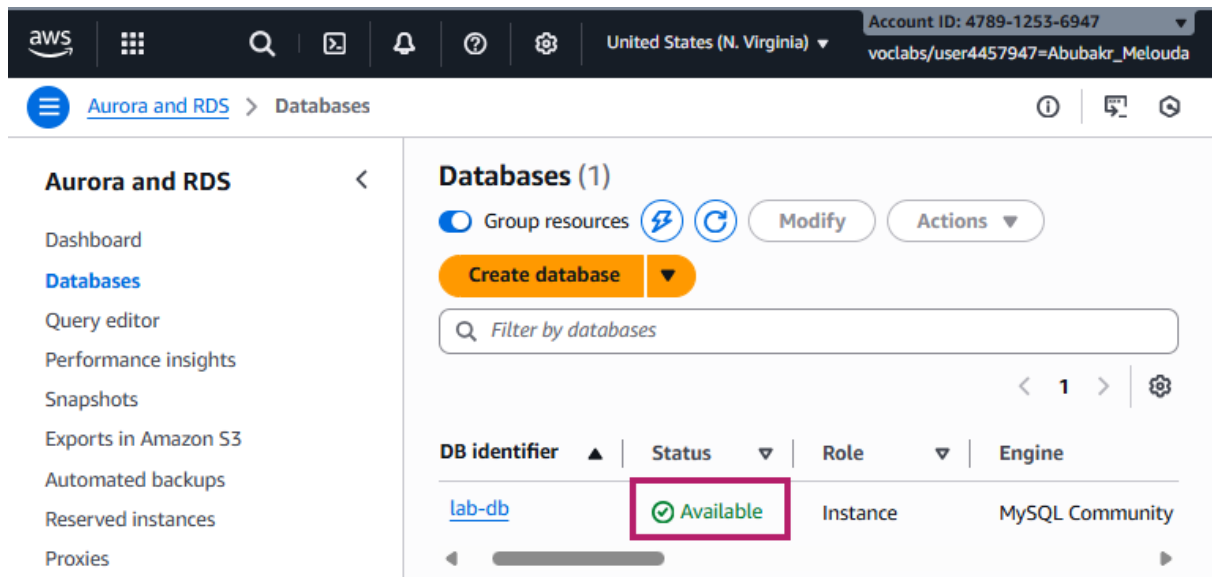
Enhanced Monitoring
☒ Enable Enhanced monitoring
Enabling Enhanced Monitoring metrics are useful when you want to see how different processes or threads use the CPU.

- Screenshot RDS database creation initiated

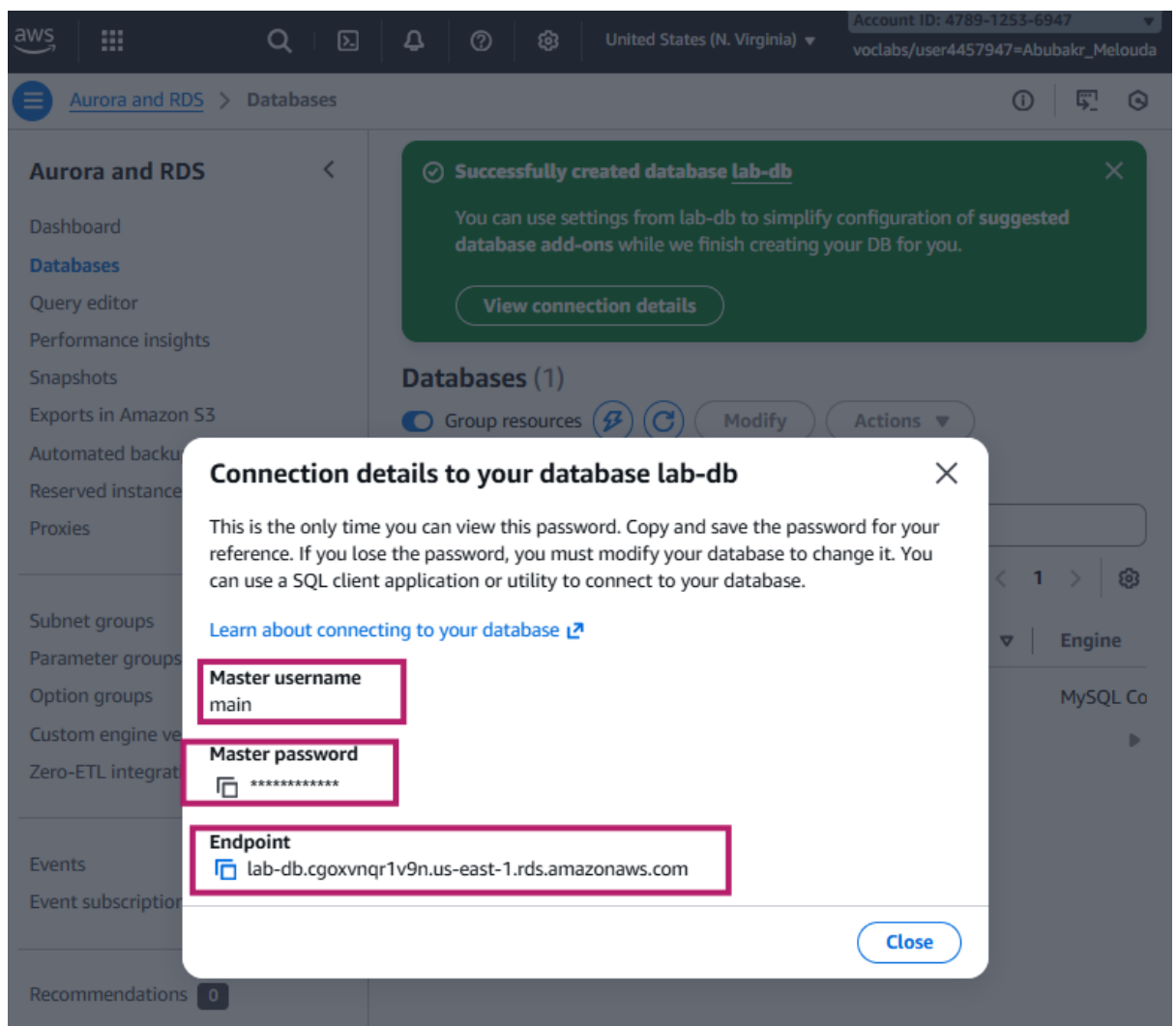


- Screenshot RDS database successfully created and available





- Screenshot database endpoint copied for application connection



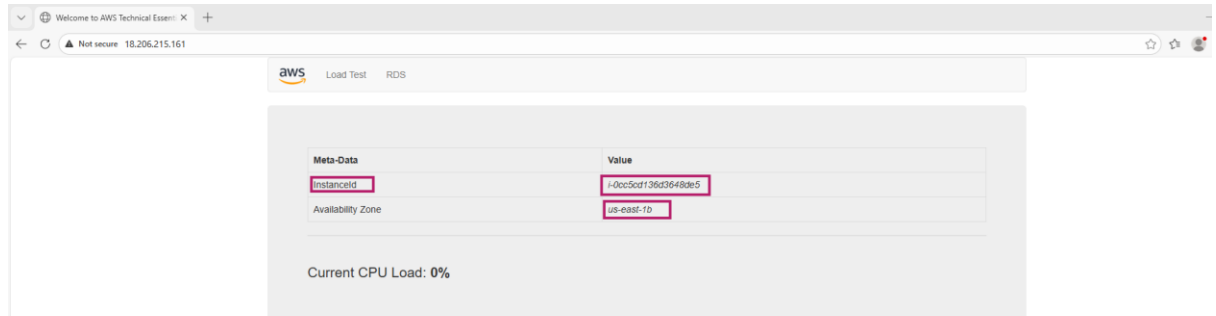
- **Screenshot web server IP address from AWS Details**

(1) ips -- public:54.235.39.166, private:10.0.0.90 (2) ips -- public:18.206.215.161, private:10.0.2.104

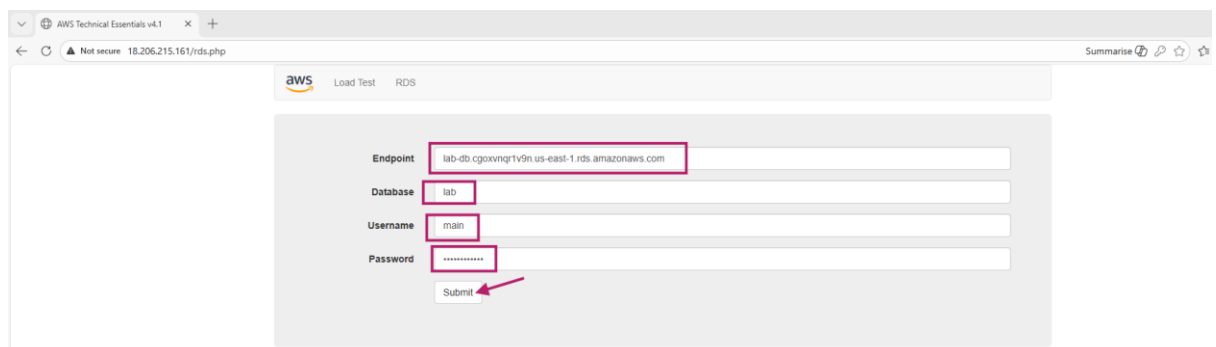
SSH key [Show](#) [Download PEM](#) [Download PPK](#)
AWS SSO [Download URL](#)

SecretKey	gjpt8BGgNnOL1qQv3WoQRd8ngJF7Q3rL2cyFjfHA
WebServer	18.206.215.161
BastionHost	54.235.39.166
Region	us-east-1
AccessKey	AKIAW7ALJMFZQ2TDDGXH

- **Screenshot web application homepage showing EC2 instance details**



- **Screenshot configuring web application to connect to RDS database**

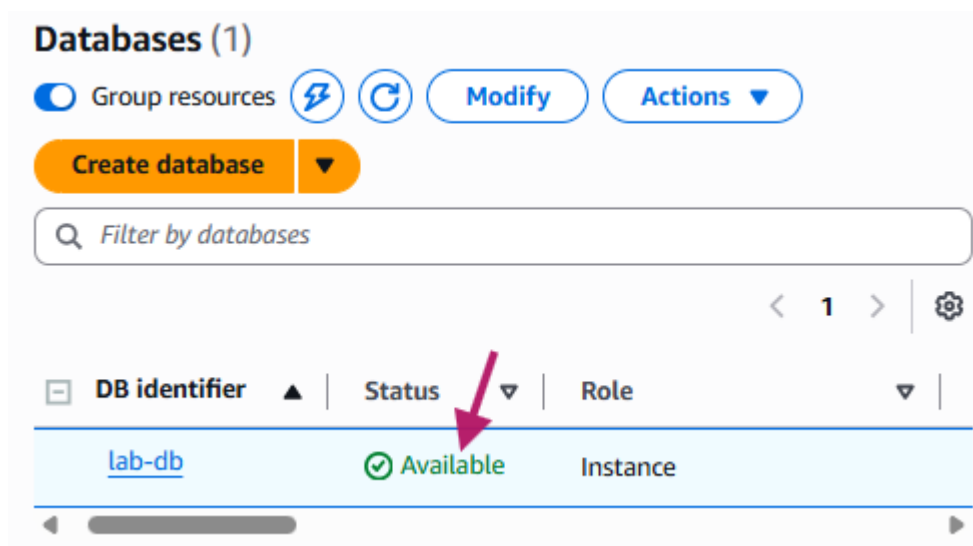


- Screenshot receiving 504 error message when submitting the login credentials to connect to the RDS database



Troubleshooting steps taken:

Verified RDS database is in available state



Upon verifying Inbound rules for DB security group, I found no MYSQL/Aurora rule set to allow access to the database.

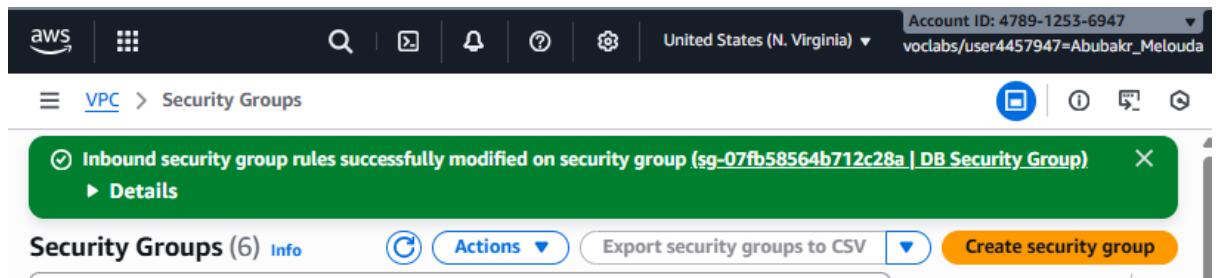
The screenshot shows the AWS Management Console interface for Security Groups. At the top, there's a navigation bar with the AWS logo, search icons, and account information (United States (N. Virginia), Account ID: 4789-1253-6947). Below the navigation bar, the 'Security Groups' page is displayed. A table lists several security groups, including 'DB Security Group' and three 'default' groups. The 'DB Security Group' is selected, and its details are shown below. The 'Inbound rules' tab is active, and a red box highlights the 'Inbound rules' section, which currently shows 'No security group rules found'.

- Screenshot adding MYSQL/Aurora inbound rule

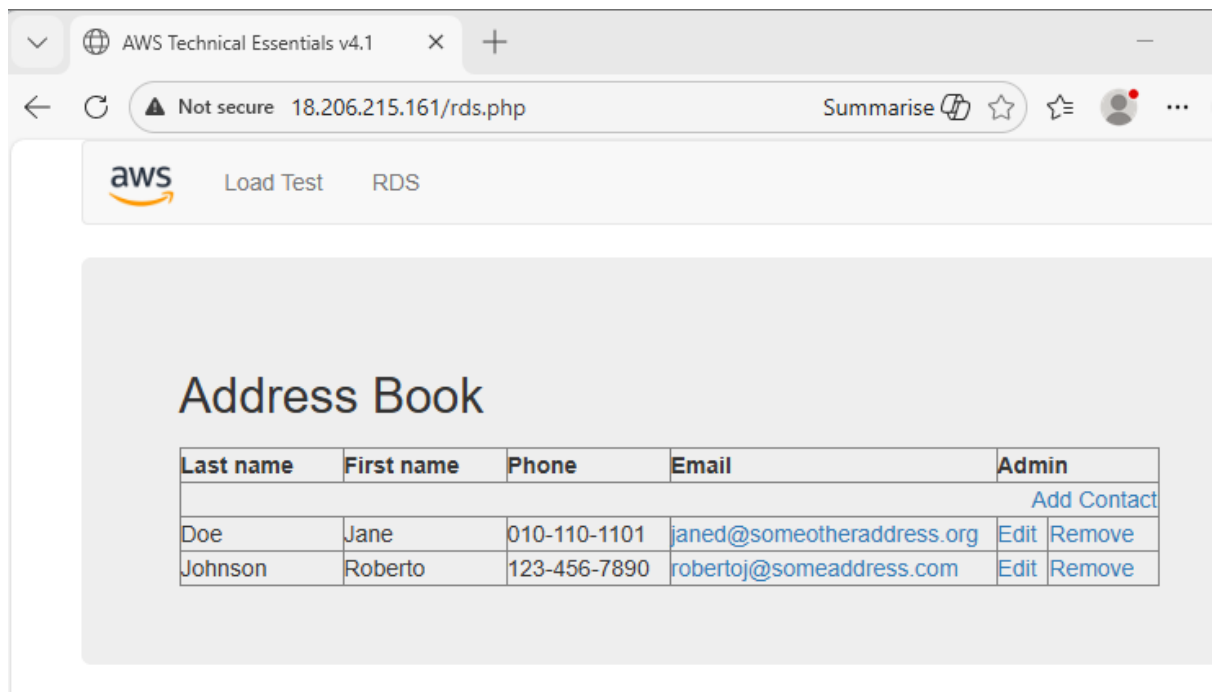
The screenshot shows the 'Edit inbound rules' page for the 'sg-07fb58564b712c28a - DB Security Group'. The page title is 'Edit inbound rules' with an 'Info' link. Below the title, a message states: 'Inbound rules control the incoming traffic that's allowed to reach the instance.' The 'Inbound rules' section shows a single rule with the following details:

- Security group rule ID:** sgr-08f93f7cc1f8e0cf4
- Type:** MYSQL/Aurora (highlighted with a red box)
- Protocol:** TCP
- Port range:** 3306
- Source type:** Anywhere-IPv4 (highlighted with a red box)
- Source:** 0.0.0.0/0 (highlighted with a red box)
- Description - optional:** MySQL/Aurora, Source: Web Security Gi

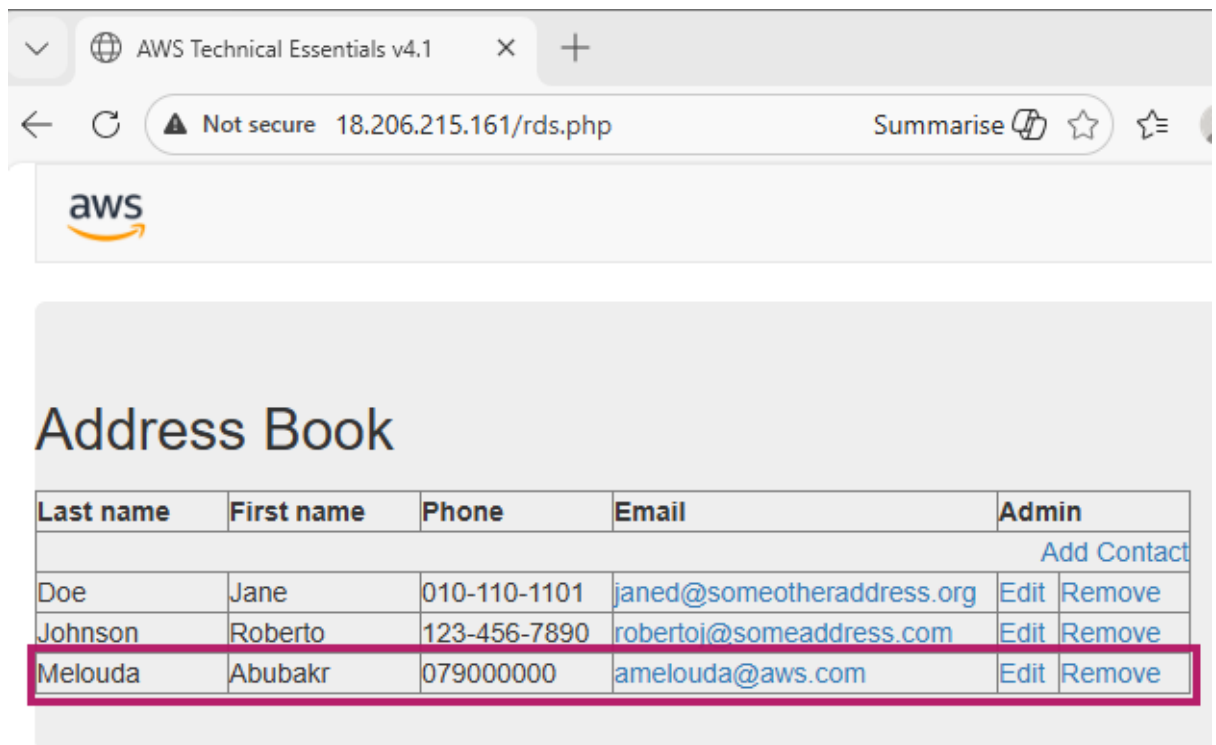
 At the bottom right, there are three buttons: 'Cancel', 'Preview changes', and 'Save rules'. A red arrow points to the 'Save rules' button.



- Screenshot address book application successfully connected to RDS database



- Screenshot test data added to database through web application



Lab summary:

Created Multi-AZ MySQL RDS database with security groups and subnet groups. Connected web application to database for data persistence.

Key Features:

- Multi-AZ deployment for high availability
- Security group configuration for database access
- DB subnet group across multiple AZs
- Web application integration
- Automated failover capability
- Database endpoint management